Coursework: Design and Evaluate Neural Networks on the Forest CoverType Dataset

SCCM6220 Machine Learning: 2026/27

Overview

In this coursework, you are asked to design, implement, evaluate and compare neural networks to classify data from the Forest CoverType dataset. You should begin with a **baseline multi-layer perceptron** (MLP), and then design a **deeper model**. The aim of this coursework is to assess your understanding of key machine learning principles, including optimization, regularization, and model evaluation.

Your submission should demonstrate your ability to:

1. Implement neural network architectures.
2. Explain and justify design choices such as activation functions, initialization and regularization.
3. Evaluate and compare model performance using suitable metrics.
4. Reflect on model behaviour and reproducibility.

Data

You must use a truncated version of the Forest CoverType dataset, which is included in the zipped folder as covtype-10.csv. This set contains observations from 581,012 trees in Roosevelt National Forest, Colorado. You can find more information about this at https://archive.ics.uci.edu/dataset/31/covertype. The original dataset includes 54 features, but this version has been truncated to 10, including:

- Elevation in metres
- Aspect in degrees azimuth
- Slope in degrees
- Horizontal and Vertical Distance to Hydrology in metres
- Horizontal Distance to Nearest Roadway in metres
- Hill shade at 9am, Noon and 3pm (index of 0-255)
- Horizontal Distance to Fire Points in metres

There are 7 types of tree represented in this dataset, indicated in the Cover_Type column as follows:

- 1: Spruce / Fir ($n = 211{,}840$)
- 2: Lodgepole Pine ($n = 283{,}301$)
- 3: Ponderosa Pine ($n = 35{,}754$)
- 4: Cottonwood / Willow ($n = 2{,}747$)
- 5: Aspen ($n = 9{,}493$)
- 6: Douglas Fir ($n = 17{,}367$)
- 7: Krummholz ($n = 20{,}510$)

Since this data is imbalanced and would take a long time to train, you must create a subset of this data for use in the coursework by:

- Selecting 5 of the classes at random
- Selecting 2500 samples of each class at random

In both of the above cases, you should use your student ID as a seed for the random number. As a result, you should have a new dataset with $5 \times 2{,}500 = 12{,}500$ samples.

You must further create training and testing subsets with an 80:20 split. The split does not have to be uniform over your classes.

Tasks

The tasks for this coursework include four components:

- Implementation of a baseline shallow MLP model. (25%)
- Implementation of a deeper MLP model. (30%)
- A short written report of 1200 words. (20%)
- A final quiz in week 10 to act as a VIVA, which is typical for assessment of research. (25%)

The code must be written in .py files. You must include a requirements.txt file and all random seeds must be fixed for reproducibility. The code must be your own and must not be taken from lab solutions, generative AI or online resources.

Task 1: Baseline Model (25%)

Implement a multilayer perceptron with

- One hidden layer containing 2 nodes.
- Reasonable choices of activation functions, which should be justified in the report. Your hidden layer and output activations do not need to be the same.
- Mean Squared Error or Cross Entropy as the loss function.

Your code must be written yourself using the libraries listed below. The model should be trained and evaluated on your version of the CoverType dataset, and you should produce graphs of the training and validation loss and accuracy for presentation in the report. Your code should produce and save these graphs. Please do not include the dataset in your submission, since it will be very large. Please assume that the dataset csv file is included in the same directory as your code.

Task 2: Deep Model (30%)

Implement a deeper MLP with 4 layers. You are free to choose the number of nodes per layer, but you should justify the number of nodes chosen in the report. You must include two of the following techniques with justification in the report:

- Dropout
- Batch Normalisation

- Gradient Clipping
- Weight Decay

Your code must be written yourself using the libraries listed below. The model should be trained and evaluated on the same dataset as your Baseline Model, and you should produce graphs of the training and validation loss and accuracy for presentation in the report. Your code should produce and save these graphs. Please do not include the dataset in your submission, since it will be very large. Please assume that the dataset csv file is included in the same directory as your code.

Your report must include a comparison of the results of your baseline and deep model along with an ablation study, demonstrating the impact of the techniques you included from the above.

Task 3: Evaluation and Report (20%)

Write a short report, no longer than 1200 words excluding references, tables, captions and equations. Your report must include references, tables and figures where appropriate and be structured as follows:

1. Introduction: Statement of our problem and a short overview of your dataset.
2. Methodology: Description of your architectures, including diagrams where appropriate, details of your training and testing setup, and your hyper-parameters. You must also describe your stopping criteria for both models and explain why it is fair.
3. Results: Your accuracy values and measures of error, loss curves and confusion matrices.
4. Discussion: Training ability; impact of regularisation, learning rate and hyper-parameter variations; suitability of your activation functions and consideration of possible alternatives; comparison of your approaches; reproducibility.
5. Conclusion: Summary of your findings.

Task 4: Quiz (25%)

There will be a quiz in the lab session of week 10, which is worth 25% of your overall mark. This is designed to act as a “VIVA” and test your core understanding of your methodology. The standard completion time allowed for the quiz will be 30 minutes with modified timings where appropriate.

Permitted Libraries

The code must be written in Python and you must implement the MLPs from scratch. You may use Pandas, Numpy, Matplotlib and Seaborn. You may also use train_test_split from sklearn.model_selection, StandardScaler from sklearn.preprocessing, accuracy_score and confusion_matrix from sklearn.metrics. If you want to use any additional functions, please check the Moodle FAQ page and, if the information is not there, please ask me.

Deadlines and Submission

The **code from Tasks 1 and 2 and the report from Task 3** must be submitted via the Moodle page submission point in the Coursework section by **Friday, 4th December 2026 at 4pm**. NB: This deadline is in **Week 9**. Your code must be submitted as a collection of .py files along with a requirements.txt file and your report must be submitted as a pdf. The code must be working and your results must be reproducible.

Marking Criteria

Marks will be distributed as follows:

Baseline Model (Task 1)	**Marks**
Model correctness and implementation, including convergence and agreement with written report	20
Code quality and reproducibility, including commenting	5
Total	**25**

Deep Model (Task 2)	
Model correctness and implementation, including convergence and agreement with written report	20
Effective implementation of the techniques (dropout etc)	5
Code quality and reproducibility, including commenting	5
Total	**30**

Report (Task 3)	
Structure, tone, clarity, presentation	4
Justification of architecture choices	4
Logical comparisons	4
Insight demonstrated about training behaviour and model components, including initialization and regularization	4
Understanding of limitations and reproducibility	4
Total	**20**

Quiz (Task 4)	
Total	**25**

Grand Total	**100**

Generative AI and Plagiarism

In the University's RAG traffic light system for Generative AI, this assessment is AMBER. This means that you can use generative AI to help you with research and to help find bugs in code. But you must not use generative AI to write your code, your report or any parts of them. You remain responsible for the creation of your submitted work and your understanding of it. Remember that the quiz in week 10 is designed to test your understanding. Further, your code must be your own and may not be copied from online repositories or libraries.